Stat 135 Section

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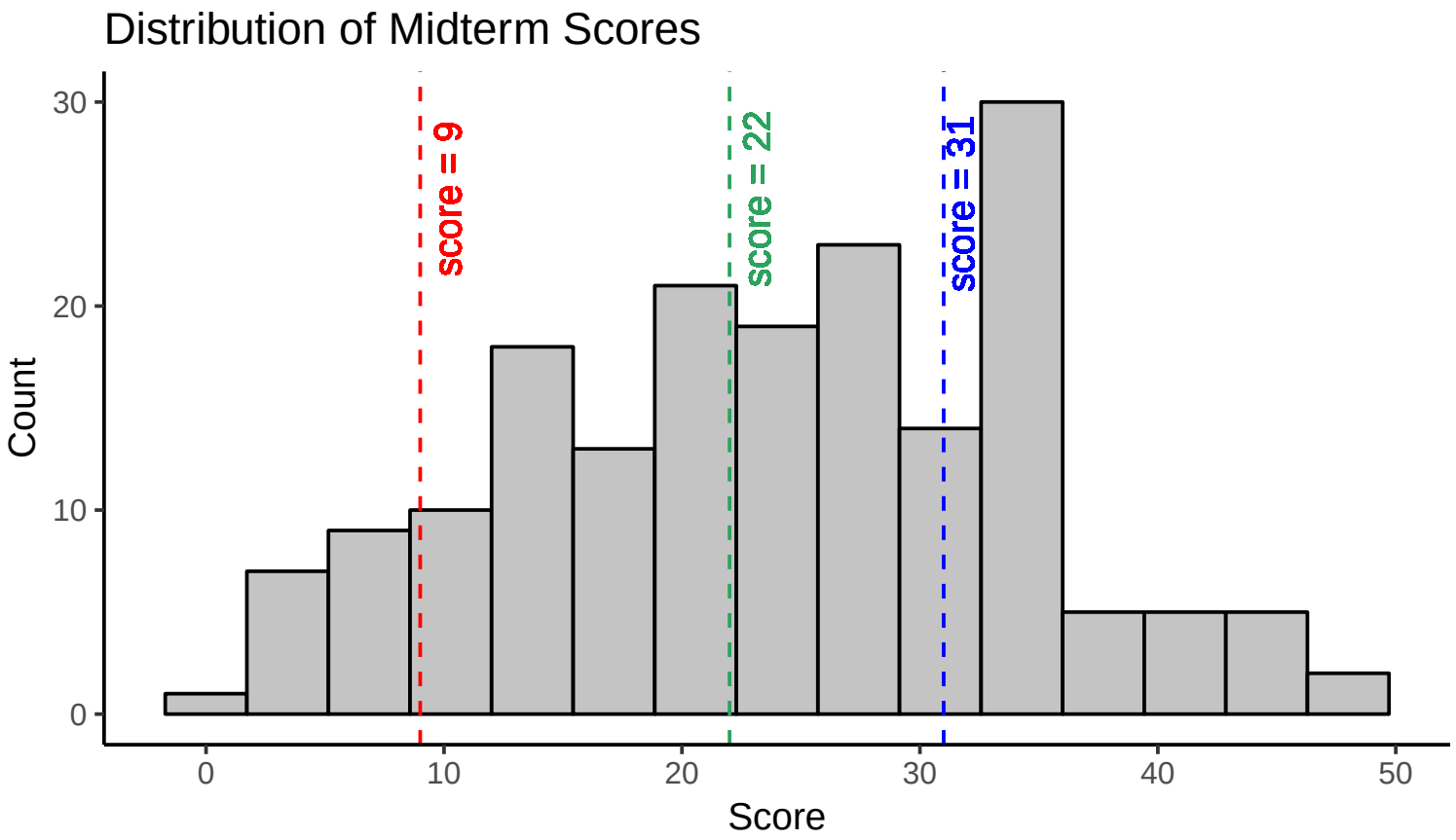
Midterm Statistics



Mean: 23.93, Median: 25, SD: 10.598

Question 1

Mean: 6.37, Median: 7, SD: 2.88

- confused between parametric and non-parametric bootstrap
- did not know how / used inappropriate methods to construct confidence intervals

Question 2

Mean: 14.45, Median: 15, SD: 6.69

- minimal sufficiency criterion is **bi-directional**, cannot just claim it from verifying

$$T(X) = T(Y) \Rightarrow p_\theta(X)/p_\theta(Y) \text{ doesn't depend on } \theta$$

  (other direction is important!)
- to show UMP, you need to **explicitly verify** that the rejection region does not change

Question 3

Mean: 3.12, Median: 2, SD: 3.38

- 3(a) equivariance does **not** mean that

$$\mathrm{Var}((\hat{\theta}_{\mathrm{MLE}})^2) = \mathrm{Var}(\hat{\theta}_{\mathrm{MLE}})$$

- 3(b) many people thought MLE achieves CRLB. This is only true **asymptotically**!
- almost no one got 3(c) (if you did, well done!)

Post-Midterm...

- Midterm will be dropped if perform better on the final
- Quiz 3 on the week after break (have more time to prepare)
- Don't give up, keep trying!

Outline

Topic: Comparing two samples

1. Overview
2. Parametric approaches
3. Non-parametric approaches
4. Examples
5. More approaches (not covered in class)

Problem 1 (11.6.48, Rice)

Proteinuria, the presence of excess protein in urine, is a symptom of renal (kidney) distress among diabetics. Taguma *et al.* (1985) studied the effects of captopril for treating proteinuria in diabetics. Urinary protein was measured for 12 patients before and after eight weeks of captopril therapy. The amounts of urinary protein (in g/24 hrs) before and after therapy are shown in the following table. What can you conclude about the effect of captopril? Consider using parametric or nonparametric methods and analyzing the data on the original scale or on a log scale.

Problem 1 Data

Before	After
24.6	10.1
17.0	5.7
16.0	5.6
10.4	3.4
8.2	6.5
7.9	0.7
8.2	6.5
7.9	0.7
5.8	6.1
5.4	4.7
5.1	2.0
4.7	2.9

Source: Rice, *Mathematical Statistics and Data Analysis* (3ed), p. 475

Problem 2 (11.6.49, Rice)

Egyptian researchers took a sample of 126 police officers subject to inhalation of vehicle exhaust in downtown Cairo and found an average blood level concentration of lead equal to 29.2 μg/dl with a standard deviation of 7.5 μg/dl. A sample of 50 policemen from a suburb, Abbasia, had an average concentration of 18.2 μg/dl and a standard deviation of 5.8 μg/dl. Form a confidence interval for the population difference and test the null hypothesis that there is no difference in the populations.

Performing parametric tests in R

- t-test (unpaired)

```
# unpaired
t.test(x, y) # x and y numeric vectors
```

- t-test (paired)

```
# paired
t.test(x, y, paired = TRUE) # x and y numeric
vectors
```

You can specify which t-test by specifying values of `var.equal` (type `?t.test` on your **R** console)

Performing non-parametric tests in R

- Wilcoxon signed rank test (paired data)

```
# dependent 2-group Wilcoxon signed rank test
wilcox.test(y1, y2, paired = TRUE) # y1 and y2
numeric vectors
```

- Wilcoxon rank sum (Mann-Whitney) test

```
# independent 2-group Mann-Whitney U Test
wilcox.test(x, y) # y and x numeric vectors
```

Chi-square tests

- General method for testing independence, homogeneity and multinomial models (GOF)
- **Read 13.3 & 13.4 of Rice!!**
- **Start on Homework 7!!**
- Next three slides provide simple examples for each scenario

1. Chi-square (GOF) test for multinomial models

When crossing certain types of red and white roses, one obtains red, white and pink roses. Theory predicts that proportion of red to white to pink roses is $3:2:2$. Test the plausibility of this theory when out of a sample of 80 crosses, 35 are red, 31 are white and 14 are pink.

	Red	White	Pink	Total
Observed	35	31	14	80
Expected	$\frac{3}{7} \times 80$	$\frac{2}{7} \times 80$	$\frac{2}{7} \times 80$	80

1. Chi-square test for multinomial models (Solution)

Recall that the chi-squared statistic X^2 for this dataset $\mathcal{D}$ is

$$\begin{aligned} X^2(\mathcal{D}) &= \sum_{j=1}^{3} \frac{(O_j - E_j)^2}{E_j} \\ &= \frac{(35 - \frac{3}{7} \times 80)^2}{\frac{3}{7} \times 80} + \frac{(31 - \frac{2}{7} \times 80)^2}{\frac{2}{7} \times 80} + \frac{(14 - \frac{2}{7} \times 80)^2}{\frac{2}{7} \times 80}. \end{aligned}$$

X^2 is approximately χ^2 distributed with $3 - 1 = 2$ degrees of freedom under the null model (which asserts ratio $3 : 2 : 2$). So we reject the null model at level α if $X^2(\mathcal{D}) > \chi^2_2(\alpha)$.

1. Chi-square test for multinomial models (Solution)

You can implement the test in **R** as follows.

```
# feed the data vector
observed <- c(35, 31, 14)
exp.prob <- c(3, 2, 2) / 7
# perform chi-square test
chisq.test(observed, exp.prob)
```

Exercise: Run the code above in RStudio.

2. (Pearson's) Chi-square test for independence

Data on the hair colour (blond, brown, dark) and eye colour (bright, dark) of 65 individuals was collected.

	Blond	Brown	Dark	Row Sums
Bright	12	8	6	26
Dark	2	25	12	39
Column Sums	14	33	18	65

Assuming the sample of individuals here is representative of a larger population, are hair and eye colour independent attributes for this population? Or are they associated with one another?

2. Chi-square test for independence (Solution)

The chi-squared statistic X^2 for this data set $\mathcal{D}$ is

$$\begin{aligned} X^2(\mathcal{D}) &= \sum_{i=1}^{2}\sum_{j=1}^{3}\frac{(O_{ij}-E_{ij})^2}{E_{ij}} \\ &= \frac{(12-\frac{14\times 26}{65})^2}{\frac{14\times 26}{65}} + \frac{(8-\frac{33\times 26}{65})^2}{\frac{33\times 26}{65}} + \frac{(6-\frac{18\times 26}{65})^2}{\frac{18\times 26}{65}} \\ &\quad + \frac{(2-\frac{14\times 39}{65})^2}{\frac{14\times 39}{65}} + \frac{(25-\frac{33\times 39}{65})^2}{\frac{33\times 39}{65}} + \frac{(12-\frac{18\times 39}{65})^2}{\frac{18\times 39}{65}}. \end{aligned}$$

X^2 is approximately χ^2 distributed with $(I-1)(J-1) = (2-1)(3-1) = 2$ degrees of freedom under the null model (which asserts independence of the eye colour and hair colour). So we reject the null if $X^2(\mathcal{D}) > \chi^2_2(\alpha)$.

2. Chi-square test for independence (Solution)

You can implement the test in **R** as follows.

```
# feed the table of observations
table <- matrix(c(12, 2, 8, 25, 6, 12), ncol = 3)
# perform chi-square test
chisq.test(table)
```

Exercise: Check that `table` *corresponds to the given table, and run the code.*

3. Chi-square test for homogeneity

We asked $n_1 = 24$ girls and $n_2 = 17$ boys how they preferred their Sunday morning breakfast egg (Sunny Side Up, Hard Boiled, or Scrambled), and collected the following table.

	Sunny	Boiled	Scrambled	Row Sums
Girl	5	12	7	24
Boy	9	3	5	17
Column Sums	14	15	12	41

Is the distribution of egg preference the same for (i.e., homogeneous across) boys and girls? Or is there a difference in the boy and girl distributions?

3. Chi-square test for homogeneity (Solution)

The chi-squared statistic X^2 for this data set $\mathcal{D}$ is

$$\begin{aligned} X^2(\mathcal{D}) &= \sum_{i=1}^{2}\sum_{j=1}^{3}\frac{(O_{ij}-E_{ij})^2}{E_{ij}} \\ &= \frac{(5-\frac{14\times 24}{41})^2}{\frac{14\times 24}{41}} + \frac{(12-\frac{15\times 24}{41})^2}{\frac{15\times 24}{41}} + \frac{(7-\frac{12\times 24}{41})^2}{\frac{12\times 24}{41}} \\ &\quad + \frac{(9-\frac{14\times 17}{41})^2}{\frac{14\times 17}{41}} + \frac{(3-\frac{15\times 17}{41})^2}{\frac{15\times 17}{41}} + \frac{(5-\frac{12\times 17}{41})^2}{\frac{12\times 17}{41}}. \end{aligned}$$

X^2 is approximately χ^2 distributed with $(I-1)(J-1) = (2-1)(3-1) = 2$ degrees of freedom under the null model (which asserts homogeneity of distributions across boys and girls). So we reject the null if $X^2(\mathcal{D}) > \chi^2_2(\alpha)$.

3. Chi-square test for homogeneity (Solution)

Exercise 1: Implement the test on your own in **R**. *Use Example 2's code as a guide if you are lost.*

Exercise 2: You should have obtained a warning message that reads "Chi-squared approximation may be incorrect" if you tried Exercise 1. Why is this so? (Answer on the next slide.)

3. Chi-square test for homogeneity (Solution)

Exercise 2. To find out what caused the error, the easiest way is to understand how the `chisq.test()` function works. Looking at the source code for `chisq.test()` (which you can access here), you will find that the warning is printed `if (any(E < 5) && is.finite(PARAMETER))`, where you can try to interpret `E` and `PARAMETER` by reading preceding lines of codes. Actually, `E` is the table of expected counts under the null model, so the warning is printed if any expected count E_{ij} is less than 5. This is consistent with the goodness of the approximate chi-square distribution of the test statistic X^2, which requires expected counts to be larger than 5. (See the second sentence on p. 519 of Rice.)

Important facts about chi-square tests for Quiz 3

- Know what each of the three tests is used for, and know when to use them
- Know how to use them
  - can write out the chi-square statistic
  - can compute degrees of freedom
- Know when the chi-square test is inappropriate (see Exercise 2 on previous slide)

Have a great Spring break!

Punggol Waterway, Singapore